

**Study and Examination Regulations of the Department of Applied Computer Science at Fulda University of Applied Sciences for the Master's degree program Global Software Development dated January 22, 2020, amended on April 14, 2021, October 19, 2022, November 16, 2022, April 12, 2023, January 17, 2024 and January 16, 2024.
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§ 1 Study objectives, study program variants, academic degree

- (1) In "Global Software Development", complex software systems are designed and implemented in a team that may be distributed around the world. The aim of the degree program is to Graduates to successfully work in and lead such teams, with a particular focus on the special challenges of international projects with team members from different cultural backgrounds,

are aware of. An essential basis for this is the personal experience of a stay abroad. In addition to technical expertise from all areas of software development, students are taught methodological and analytical principles and their application in an intercultural team. The course should enable students to solve problems independently, even in new and unfamiliar situations, to recognize and independently adapt recurring concepts and to carry out and reflect on their actions in a methodically well thought-out manner.

- (2) The Master's examination of Global Software Development degree program is designed to ensure that Graduates have acquired in-depth specialist knowledge, can successfully work on both practical and fundamental problems and have the ability to develop scientific methods and findings. They will be able to select, critically interpret and apply current research findings relevant to their work.
- (3) The Master's degree course in Global Software Development is available in two program variants with different standard periods of study:
 - as a three-semester course variant for students with a qualifying university degree with a total of 210 ECTS credits in accordance with § 2 Para. 1 No. 1a)
 - as a four-semester course variant for students with a qualifying university degree with a total of 180 ECTS credits in accordance with § 2 Para. 1 No. 1b).
- (4) Upon successful completion of the degree program, Fulda University of Applied Sciences awards the academic degree "Master of Science" (abbreviation: "M. Sc.").

§ 2 Admission requirements, admission

- (1) Admission requirements are:

A degree in computer science or a computer science-related subject from a German university or a comparable foreign university.

According to German standards, the scope of these studies must be at least

- a) six semesters and 180 ECTS credits or
- b) seven semesters and 210 ECTS credits

i.e. applicants with a foreign degree must provide evidence of a degree equivalent to a German degree of at least three years (para. 1a) or three and a half years (para. 1b).

The grade point average must be at least 2.5 or better or 75% of the maximum achievable grades.

Successful proof of professional aptitude is provided as part of an aptitude test conducted by the Department of Applied Computer Science. This specialist knowledge includes sufficient skills from the field of applied computer science such as

B. programming, algorithms and data structures, software engineering and databases as well as from application areas such as communication systems, media informatics, operational applications or embedded systems. This aptitude test may be waived for applicants who have a university degree from a country in which the ECTS standards apply. The decision on this is made by the head of degree program.

Qualified English language skills must be demonstrated, usually by a TOEFL test (computer-based at least 213 points, paper-based at least 550 points, internet-based at least 85 points). Alternatively, an IELTS test with a score of band 6.5 or higher is also accepted.

Applicants who not obtained their higher education entrance qualification at a German-speaking institution must provide proof of qualified German language skills at level A1 of the Common European Framework of Reference for Languages, usually in the form of a Goethe-Zertifikat A1, telc-Zertifikat A1, ÖSD-Zertifikat A1 or ECL-Zertifikat A1. Alternatively, the language certificates DSH1 or TestDaF at least TDN 3 are also accepted.

- (2) Admission to the course is possible in the summer and winter semesters. If you are admitted in the winter semester, your studies start with the modules of the second semester and then continue in the summer semester with the modules of the first semester.

§ 3 Application documents

The following documents must be submitted with your application:

- Curriculum vitae in tabular form on one page including a passport photo (4cm x 5 cm)
- Proof of English language competence
- Proof of German language skills
- School leaving certificate from secondary school
- Certificate of the academic degree obtained in the first degree program
- Degree certificate from the first degree program with a complete list of all subjects studied and their grades (including a description of the grading system used by the university).

§ 4 Standard period of study, ECTS credits of the degree program

- (1) The standard period of study for the three-semester course variant is three semesters; a total of 90 ECTS credits must be earned.
- (2) The standard period of study for the four-semester course variant is four semesters; a total of 120 ECTS credits must be earned.

§ 5 Modules

- (1) The program includes the modules Cloud Computing (AI5029), Big Data Technologies (AI5025), Model-Driven Development (AI5084), Reconfigurable Computing (AI5100), Programming Paradigms (AI5055), Building Web and Mobile Apps (AI5036), Distributed Applications (AI5109), Global Distributed Software Development (AI5088), Machine Learning (AI5031), Parallel Programming (AI5085), Test Oriented Development (AI5086), Intercultural Communication and Scientific Work (SK5900), Master's Thesis in Global Software Development (AI5117) and a compulsory elective module (see section 3).

In addition, two of the above modules except Intercultural Communication and Scientific Work (SK5900) and the thesis module (AI5117) can each be replaced by another module from the compulsory elective catalog (see paragraph 3).

- (2) Students of the four-semester course variant who are not obliged to study abroad in accordance with § 6 must also take one of the following modules: Additional Courses (AI4001), Master Internship (AI5119) or German Language (AI5118).
- (a) Additional Courses (AI4001):
As part of the Additional Courses module (AI4001), students must take modules from the degree programs of the Department of Applied Computer Science amounting to 30 ECTS credits, which represent a meaningful addition to their previous studies with regard to the degree they are aiming for. Sufficient knowledge of German (DSH2) is required for this. The subjects are determined in consultation with the head of degree program.²
- (b) Master Internship (AI5119):
Students who do not meet the requirements for the module Additional Courses (AI4001) because they do not have the necessary knowledge of German can alternatively complete the module Master Internship (AI5119) in the third semester at a location of a company, organization or research institution within Germany. The prerequisite for admission to the Master Internship module (AI5119) is that modules worth 40 ECTS points have already been successfully completed. The duration of the Master Internship module (AI5119) is usually 20 weeks. During this time, students must work the normal working hours of full-time employees. Missed practical periods must be made up. The details are regulated in the work placement regulations (Annex 3).
- (c) German Language (AI5118):
Students who do not meet the requirements for the modules Additional courses (AI4001) and Master Internship (AI5119) because they lack the necessary German language skills can have these language skills recognized through language courses amounting to 30 ECTS as part of the module German Language (AI5118).
- (d) A combination of the alternatives mentioned in paragraphs (b) and (c) is excluded.
- (3) The module Current Topic of Global Software Development (AI5095) can be taken several times as a compulsory elective module if it has been taken with different content.
- (4) The structure of the curriculum is shown in Appendix 1.
- (5) Students choose a module from the following catalog of compulsory elective modules:
- Intercultural Project Management (SK5901)
 - Software Development Project (AI5094)
 - Research Project (AI5016)
 - Current Topic of Global Software Development (AI5095)
 - Advanced Big Data Big Data (AI5122)
 - Cryptography - Theory and Applications (AI5046)
 - User Centered Development (AI5083)
 - Stream Processing and Analytics (AI5174)

§ 6 Study abroad

- (1) All students who have not already completed at least one semester of study abroad during their Bachelor's degree course or who have not completed the Bachelor's degree on which admission is based at a foreign university.

have completed a semester abroad in a foreign language according to the following conditions:

- (a) Students on the three-semester degree program usually complete a semester abroad in their regular second or third semester.
 - (b) Students on the four-semester course usually complete a semester abroad in the third semester by taking the module Intercultural Study of Computer Science (AI5092) or the module Master Internship (AI5119) as an internship abroad.
- (2) Further details are regulated in the separate regulations of the Department of Applied Computer Science for studying abroad.

§ 7 Thesis module Master's Thesis in Global Software Development

- (1) The module Master's Thesis in Global Software Development (AI5117) consists of the master's thesis and a colloquium.
- (2) The Master's thesis takes six months to complete. It can be extended once by up to four weeks by the first examiner at the student's request.
- (3) The first examiner of the thesis must be a professor in the Department of Applied Computer Science.

§ 8 Grading of the modules

- (1) All modules are graded with the exception of the modules Intercultural Study of Computer Science (AI5092), Additional Courses (AI4001), German Language (AI5118) and Master-Internship (AI5119).
- (2) In the module Master's Thesis in Global Software Development (AI5117), the colloquium is not graded. The module grade corresponds to the grade of the Master's thesis.
- (3) If the modules Intercultural Study of Computer Science (AI5092), Additional Courses (AI4001), German Language (AI5118) or Master-Internship (AI5119) as well as the colloquium of the module Master's Thesis in Global Software Development (AI5117) are successfully completed, they will each receive the grade "successfully completed".

§ 9 Free attempt, grade improvement, crediting of examination attempts

- (1) Up to two module examinations that students complete within their first two semesters can either be counted as not taken if they have not been passed for the first time (free attempt) or can be repeated once to improve the grade if they have passed the examination (grade improvement). The better result counts.

§ Section 20 (3) ABPO 2018 applies accordingly. The module Master's Thesis in Global Software Development (AI5117) is excluded from this.

- (2) Failed attempts and passed coursework and examinations for identical modules from other degree programs will be credited.

§ 10 Formation of the overall grade

- (1) The degree program is successfully completed if all modules required for the degree have been graded at least "sufficient".

- (2) The overall grade is the arithmetic mean of the module grades weighted according to ECTS points.

§ 11 Entry into force, transitional rule

- (1) These examination regulations come into force on 01.10.2020.
- (2) Students who were already enrolled in the degree course "Global Software Development" before these examination regulations came into force will complete their studies in accordance with the previous examination regulations dated 17.10.2018, amended on 22.01.2020. This option will end at the end of the winter semester 2022/2023. After that, students will receive recognition of the work already completed in accordance with a recognition rule adopted by the Faculty Council. A voluntary change to these examination regulations is possible upon request.

Appendix 1: Structure of the curriculum

Study plan for students in the three-semester course variant, module overview with examination forms

Module ID	Module	Form of examination
	1st semester (summer)	
AI5029	Cloud computing	Elaboration or written exam
AI5025	BigData Technologies	Written exam or portfolio
AI5084	Model-Driven Development	Portfolio or written exam
AI5100	Reconfigurable Computing	Written exam or technical discussion
AI5055	Programming Paradigms	Written exam or term paper
	Compulsory elective module	
	• Intercultural Project Management (SK5901)	Project work or presentation
	• Software Development Project (AI5094)	Presentation
	• Current Topic of Global Software Development (AI5095)	Project work or written exam or presentation
	• Research Project (AI5016)	Project work or portfolio
	• Advanced Big Data (AI5122)	development
	• Cryptography - Theory and Applications (AI5046)	Written exam or technical discussion
	• User Centered Development (AI5083)	Portfolio
	• Stream Processing and Analytics (AI5174)	Project work or portfolio
	2nd semester (winter)	
AI5036	Building Web and Mobile Apps	Portfolio
AI5109	Distributed Applications	Written exam or technical discussion
AI5088	Global Distributed Software Development	Project work or presentation
AI5031	Machine Learning	Exam
AI5085	Parallel Programming	Term paper or technical discussion
AI5086	Test Oriented Development	Term paper or written exam

Module ID	Module	Form of examination
	3rd semester (summer)	
AI5117	Master's Thesis in Global Software Development	Term paper and colloquium
SK5900	Intercultural Communication and Scientific Work	Portfolio or presentation

Study plan for students in the four-semester course variant, module overview with examination forms

Module ID	Modules	Form of examination
	1st semester (summer)	
AI5029	Cloud computing	Elaboration or written exam
AI5025	BigData Technologies	Written exam or portfolio
AI5084	Model-Driven Development	Portfolio or written exam
AI5100	Reconfigurable Computing	Written exam or technical discussion
AI5055	Programming Paradigms	Written exam or term paper
	Compulsory elective module	
	Intercultural Project Management (SK5901)	Project work or presentation
	Software Development Project (AI5094)	Unit
	Current Topic of Global Software Development (AI5095)	Project work or written exam or presentation
	Research project (AI5016)	Project work or portfolio
	Advanced Big Data (AI5122)	development
	Cryptography - Theory and Applications (AI5046)	Written exam or technical discussion
	User Centered Development (AI5083)	Portfolio
	Stream Processing and Analytics (AI5174)	Project work or portfolio
	2nd semester (winter)	
AI5036	Building Web and Mobile Apps	Portfolio
AI5109	Distributed Applications	Written exam or technical discussion
AI5088	Global Distributed Software Development	Project work or presentation
AI5031	Machine Learning	Exam
AI5085	Parallel Programming	Term paper or technical discussion
AI5086	Test Oriented Development	Term paper or written exam

Module ID	Modules	Form of examination
	3rd/4th semester (summer)	
	In accordance with § 5 (2), § 6 (1): • Additional Courses (AI4001) or • Master Internship (AI5119) (3rd semester) or • German Language (AI5118) or • Intercultural Study of Computer Science (AI5092)	See module description of the selected modules Report By offer The forms of examination depend on the modules of the foreign university
	3rd/4th semester (summer)	
AI5117	Master's Thesis in Global Software Development	Term paper and colloquium
SK5900	Intercultural Communication and Scientific Work	Portfolio or presentation

Overview of module allocation/ semester abroad for three- and four-semester course variants

	four-semester course variant	three-semester course variant
Students with a German bachelor's degree+ existing Semester abroad	Additional Courses (AI4001)	-
Students with a German bachelor's degree without a semester abroad	Intercultural Studies of Computer Science (AI5092) or Master Internship (AI5119) abroad	Semester abroad (recognition at module level)
Students with an international Bachelor's degree	Additional Courses (AI4001) <u>or</u> Master Internship in Germany (AI5119) <u>or</u> German Language (AI5118)	-

Appendix 2: Module descriptions

Compulsory modules:	12
AI5029 Cloud Computing	12
AI5025 Big Data Technologies	14
AI5084 Model-Driven Development	16
AI5100 Reconfigurable Computing	17
AI5055 Programming Paradigms	19
AI5036 Building Web and Mobile Apps	20
AI5109 Distributed Applications	22
AI5088 Global Distributed Software Development	23
AI5031 Machine Learning	24
AI5085 Parallel Programming	26
AI5086 Test Oriented Development	27
AI5117 Master's Thesis in Global Software Development	29
SK5900 Intercultural Communication and Scientific Work	30
Compulsory elective modules	31
SK5901 Intercultural Project Management	31
AI5094 Software Development Project	32
AI5095 Current Topic of Global Software Development	33
AI5016 Research Project	34
AI5122 Advanced Big Data	36
AI5046 Cryptography - Theory and Applications	38
AI5083 User-Centered Development	39
AI5174 Stream Processing and Analytics	41
Modules for the four-semester variant:	43
AI4001 Additional Courses	43
AI5119 Master Internship	44
AI5118 German Language	45
AI5092 Intercultural Study of Computer Science	46

Compulsory modules:

AI5029 Cloud Computing				
Module code FB:		English module name: Cloud computing		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 1. Semester AIM 2017 & 2024: 2. Semester	Frequency of the course: Summer semester	Duration: 1 semester
Art: GSD: Compulsory module AIM 2024: Compulsory elective module (IT infrastructure) AIM 2017: Compulsory module (Internet Engineering)	Level: Master	Applicability of the module:		
1	Qualification goals: The students • name the basics of cloud services, required virtualization solutions and their provision. • use the terminology and associated models of cloud computing correctly when realizing Internet services. • describe the characteristics of cloud computing and the associated IT infrastructure using given examples. • can professionally apply given requirements when implementing cloud-native services. • assess cloud-based services against the background of flexible provisioning, scalability, security and data protection. • analyze the costs of using cloud services in their own projects and • design suitable IT sourcing models. • evaluate the current state of research in this area and • are actively in applied research in the field of cloud computing.			
2	Contents of the module • Basics: history of cloud computing, requirements, virtualization • Cloud computing: taxonomy, evolution, characteristics • Service models: IaaS, PaaS, SaaS • Deployment models: public cloud, private cloud • Reference architectures: Technical realization, layer models • Use cases and practical implementation: compute/storage/network cloud, apps • Migration and integration strategies: cloud federation, hybrid/multi cloud • Limits of the cloud: risks, security, data protection Optional: • Cloud computing from the user and operator perspective: IT sourcing, costs/benefits • Outlook: Future developments, alternatives			
3	Teaching and learning methods: 2 SWS Seminar 2 SWS Internship			
4	Language: German or English			

5	Prerequisites for participation in the module: necessary: none recommended: Programming, computer networks, virtualization techniques and operating systems
6	Form of the examination: Elaboration or written exam
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

AI5025 Big Data Technologies				
Module code FB:		English module name: Big Data Technologies		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: DS 2024: 2nd semester GSD 2020: 1st semester AIM 2024: 1st/2nd/3rd semester AIM 2017: 1st semester	Frequency of the course: Winter and summer semesters	Duration: 1 semester
Art: DS, GSD: compulsory module AIM 2024: Compulsory elective module (economics business informatics) AIM 2017: Compulsory module (economics business informatics)	Level: Master	Applicability of the module:		
1	Qualification goals: Substantial advances in computing capacity have once again made data and data processing a core topic in computer science. The size and complexity of the data generated in business and industry today require an advanced or modified approach: Big Data and the associated technologies and strategies. After successfully completing this course, students will be able to, • specify and describe further functions of relational databases. • describe and describe the inherent restrictions of established relational database systems. • work on further applications and use cases using individual examples and derive architectural approaches from the big data use cases. • distinguish and apply the basic components of Hadoop and MapReduce hands-on. • Assess and evaluate BigData / noSQL use cases. • to indicate the challenges of administering big data solutions. • describe and discuss newer and further approaches in the field of Big Data			
2	Contents of the module • Advanced functions of relational databases (transactions, consistency, backup & recovery, replication) • Further motivation (requirements, environment, use cases) • Hierarchical and unstructured data • Basic concepts / architecture (scalability, failover, etc.) • Hadoop basics: HDFS • MapReduce and programming (hands-on) • Hadoop integration • Enterprise environment: Scoop, Flume etc. • Hadoop Scripting: Hive (& Pig) • Hadoop Administration (Oozie, Zookeeper, Cloudera / Hortonworks) • Newer tools in the Hadoop ecosystem (Tableau, Pentaho etc.) • Further approaches: XML databases (MarkLogic), time series (Influx), graph databases (Neo4J), Exadata			

3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship
4	Language: German or English
5	Prerequisites for participation in the module: necessary: none recommended: Database technologies
6	Form of the examination: Written exam or portfolio
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

AI5084 Model-Driven Development				
Module code FB:		English module name: Model-Driven Development		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: GSD: Summer semester	Duration: 1 semester
Art: GSD: compulsory module	Level: Master	Applicability of the module:		
1	Qualification goals: Students understand the process of model-driven development. This includes in particular the development and adaptation of appropriate tools. They follow the historical paradigm shift: from the idea that everything is an object to the more abstract idea that everything is a model. You can identify areas of application for model-driven development. In detail, they are able to, • explain the importance of the precision and completeness of the models, • work with and define domain-specific languages, • model transformations and code generators, • apply and implement refactorings throughout the entire development process, • use complex language specifications as a reference work.			
2	Contents of the module • Unified Modeling Language (UML) and Object Constraint Language (OCL) • Concepts for the domain-specific adaptation of a modeling language (e.g. UML profiles) • Definition of concrete and abstract syntax of visual and textual domain-specific languages (metamodeling and EBNF) • Concepts and frameworks for implementing model transformations • Concepts and frameworks for code generation • Use and implementation of refactorings at the level of constraints, models and code • Standards for the exchange and serialization of models • Possible applications of model-driven development (e.g. in application development for re-configurable hardware)			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Experience in modeling (UML, E/R or process modeling), basic knowledge the field of formal languages, good knowledge of object-oriented programming in Java and object-oriented design.			
6	Form of the examination: Portfolio or written exam			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5100 Reconfigurable Computing				
Module code FB:		English module name: Reconfigurable Computing		
Workload: 150 h, thereof 54 h attendance time 96 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester AIM 2017 & 2024: 2. Semester	Frequency of the course: Summer semester	Duration: 1 semester
Art: GSD: compulsory module AIM: Compulsory elective module (Embedded Systems)	Level: Master	Applicability of the module:		
1	Qualification goals: The students - analyze computationally intensive applications with regard to their parallelization and acceleration on reconfigurable computing systems, both in embedded systems and in data centers. - assess the structure and limitations of classical computer systems. - evaluate the design of heterogeneous reconfigurable computing systems with CPU and FPGA components. - apply algorithms of high-level synthesis (HLS) and hardware/software code design. - analyze a given problem in order to break it down into hardware and software components. - design simple applications for reconfigurable computing systems using HLS. - implement the communication interfaces between hardware and software. - analyze the design space of the HLS to meet given targets.			
2	Contents of the module - Construction of modern reconfigurable computing systems on modern system-on-chips (SoCs), which contain both CPUs and FPGAs - FPGA design using high-level synthesis (HLS) - Important algorithms of HLS (e.g. scheduling, allocation, binding) - System design using hardware/software codesign - Important hardware/software code design algorithms - HLS design of concrete applications using the example of C/C++ Optional: - Design space exploration - Implementation of your own project using hardware/software codesign and HLS			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: none			
6	Form of the examination: Written exam or technical discussion			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			

9	Remarks: none
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translation; legally non-binding

AI5055 Programming Paradigms				
Module code FB:		English module name: Programming Paradigms		
Workload: 150 h, thereof 54 h attendance time 96 h self-study		ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: Summer semester
Art: GSD: compulsory module		Level: Master	Applicability of the module:	
1	Qualification goals: The students - differentiate between programming paradigms and make a well-founded selection for a given problem. - identify and use the different types of type systems, polymorphism and functional constructs in new programming languages. - implement an interpreter for a simple functional language (especially polymeric type inference). - explain basic concepts of logic programming and use them to solve simple logic problems (e.g. Sudoku).			
2	Contents of the module - Functional programming concepts - Recursion versus iteration - Different types of polymorphism (parametric, subtype, ad-hoc) - Continuation passing style - Monads - Hindley-Milner type reference - Implementation of an interpreter for a small functional language - Unification - Logical programming - Backtracking, cutpoints			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Exercise			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Object-oriented programming, Procedural programming			
6	Form of the examination: Exam			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5036 Building Web and Mobile Apps				
Module code FB:		English module name: Building Web and Mobile Apps		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 2nd semester AIM 2024: 1st/3rd semester AIM 2017: 3rd semester	Frequency of the course: Winter semester	Duration: 1 semester
Art: GSD: compulsory module AIM: Compulsory elective module (Media Informatics)	Level: Master	Applicability of the module:		
1	Qualification goals: The students - develop solution concepts for current problems with application architecture, data model, component communication and security precautions as mobile and/or web applications and can justify the underlying decisions. - use cross-platform tools and frameworks to prototype modern mobile and web-based applications. - name the advantages and disadvantages of the various technical approaches from native development for each target platform to the exclusive use of web technologies and consider and justify these for given application scenarios. - document the development process including reflection on the basis of given framework conditions objectively and using references. - take into account guidelines for teamwork and react appreciatively and constructively to challenges that arise in teamwork and task distribution (optional, only when working in teams).			
2	Contents of the module - Application architectures for web and mobile applications - Backend development (concepts and frameworks), for example - Web server, dynamic execution, threads and performance, delivery of the front end, - Routes and APIs - Authentication and authorization - Frontend development (concepts and frameworks), for example - Component-based development, state management - Hybrid apps, runtime interpreter, cross-compiler - Single page applications and progressive web apps - Similarities and differences between the target platforms (e.g. iOS, Android, browser) - Use of peripherals and sensors in web applications and mobile applications Optional: - Testing and debugging with emulators - Project management and teamwork			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none			

	Recommended: in-depth programming knowledge of web technologies (especially ECMAScript, HTML, CSS), client-server technology, network protocols, IT security, IT project management, teamwork
6	Form of the examination: Portfolio or project work
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

AI5109 Distributed Applications				
Module code FB:		English module name: Distributed Applications		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 2nd semester AIM 2024: 1st/3rd semester	Frequency of the course: Winter semester	Duration: 1 semester
Art: GSD 2020: compulsory module AIM 2024: Compulsory elective module (IT infrastructure)	Level: Master	Applicability of the module:		
1	Qualification goals: The students • name standard platforms for distributed software systems, in particular distributed web applications. • name basic application architectures and analyze the requirements of complex distributed systems and the associated role of approaches such as middleware, microservices, SOA and EAI. • professionally implement independent distributed applications using various techniques and programming languages, in particular for server-side development.			
2	Contents of the module • Architectures of distributed Internet and web-based applications • Tasks, goals and functions of approaches such as middleware and microservices • Current concepts for distributed applications such as different middleware approaches (e.g. distributed object models, messaging, queuing, publish/subscribe, peer-to-peer) and communication paradigms (e.g. request/reply, multicast communication). • Web services, microservices and service-oriented architecture • Modern programming languages for server-side development Optional: • Frameworks and design patterns for distributed applications • Application server, container platforms/orchestration			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none Recommended: Good programming skills in Java or C, software engineering basics			
6	Form of the examination: Written exam or technical discussion			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination, active participation in the practical course			
9	Remarks: none			

AI5088 Global Distributed Software Development				
Module code FB:		English module name: Global Distributed Software Development		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 2nd semester	Frequency of the course: Winter semester	Duration: 1 semester
Art: GSD: compulsory module	Level: Master	Applicability of the module:		
1	Qualification goals: After completing this module, students will have a sound knowledge of how to design and implement a distributed (web) application in a global setting. They understand how to work with team members from different cultural backgrounds who in different locations and time zones. Students know how to apply software engineering and programming techniques to create applications that meet customer requirements. They know current tools and techniques that support distributed development in teams.			
2	Contents of the module Students will carry out a group project to learn about and implement important aspects of software engineering in an environment that simulates globally distributed software development companies. You will act in the typical roles of developer, technical lead and manager. The module offers a comprehensive and extended overview of practical methods and tools of software engineering as well as its organizational, team-oriented and communicative aspects. Special emphasis is placed on iterative, incremental, agile and user-centered design methods as well as on global software engineering, where teams are located in geographically and culturally dispersed areas. The basics of intellectual property, licensing, digital rights management, copyright and software development ethics are discussed. This project provides an integrative experience in software project organization and teamwork, complementing students' foundational skills in computer science.			
3	Teaching and learning methods: 4 SWS Internship Project with accompanying practical work in a laboratory; presentations by guests and students			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none Recommended: good programming skills in Java or Python, knowledge of JavaScript and HTML			
6	Form of the examination: Project work or presentation			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5031 Machine Learning				
Module code FB:		English module name: Machine Learning		
Workload: 150 h, thereof 54 h attendance time 96 h self-study	ECTS points: 5 ECTS	Semester of study: DS 2024: 1. Semester GSD 2020: 2. Semester AIM 2024: 1st/3rd semester AIM 2017: 3rd semester	Frequency of the course: Winter semester	Duration: 1 semester
Art: DS, GSD: compulsory module, AIM (2024); Compulsory elective module AIM (2017): Compulsory module (Data Science)	Level: Master	Applicability of the module:		
1	Qualification goals: The students - describe the mathematical and conceptual foundations of the statistical theory of learning. - describe several common learning algorithms and apply them in practice (using suitable standard libraries). - name the challenges that arise in the practice of object and pattern recognition, explain the most important solution approaches and apply these approaches to solve given tasks. - understand current scientific developments and applications in the field of deep learning.			
2	Contents of the module - Reflections and historical overview on the topic of machine intelligence - Mathematical and conceptual foundations of machine learning - Gradient descent as a basic learning method - Linear classification - Neural networks, deep learning - Elements of discrete probability theory - Evaluation measures for ML applications - Current topics in machine learning			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none Recommended: good knowledge of C/C++			
6	Form of the examination: Written exam or portfolio			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination and regular submission of exercises			

9	Remarks: none
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translation; legally non-binding

AI5085 Parallel Programming				
Module code FB:		English module name: Parallel Programming		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: AIM 2017 & 2024:2. Semester GSD 2020: 2nd semester	Frequency of the course: Winter and summer semesters	Duration: 1 semester
Art: AIM 2017, GSD: compulsory module AIM 2024: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: The students - name the techniques for programming parallel processes on multicore / multiprocessor systems and in heterogeneous, distributed systems. - independently develop solutions to given problems. - implement applications in the C programming language under Linux. develop parallel programs with OpenMP and MPI.			
2	Contents of the module Today's problems (weather or earthquake forecasts, crash test simulations, development of new drugs or chemical compounds, search engines of web service providers, ...) require enormous computing and storage power, which can generally only be provided by supercomputers and/or networked computers. In order to use this hardware efficiently, parallelized programs must be used. This module teaches some techniques for creating parallel programs. - Basics and concepts - OpenMP - Message Passing Interface (MPI) - Functional programming Optional: - GPGPU			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Practical course Seminar-based teaching with integrated exercises and practical work in the laboratory			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none Recommended: very good knowledge of C programming, knowledge of Linux			
6	Form of the examination: Term paper or technical discussion			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5086 Test Oriented Development				
Module code FB:		English module name: Test Oriented Development		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 2nd semester AIM 2024: 1st/3rd semester AI 2017: 3rd semester	Frequency of the course: Winter semester	Duration: 1 semester
Art: GSD: Compulsory module AIM: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: The students - define and differentiate technical terms of test-oriented development according to international standards. - use the covered test case development and test management concepts professionally, are familiar with them in given examples and can analyze and name faulty applications. - analyze given software quality problems for relevant components in the software life cycle, derive solutions and implement them as a test strategy. - explain how their own solutions work and argue the underlying decisions for alternative solutions. - create solutions to given problems in groups, actively contribute to the development of solutions and structure the collaborative work themselves.			
2	Contents of the module The success of software development projects depends on the selection of a suitable methodology for the creation of the software and high quality through static and dynamic tests, the models (in model-driven environments) and the usability of the HCI. This course focuses on the testing methods required in traditional (heavy-weight) development environments with test levels (from-unit-to-acceptance testing) and agile development environments with continuous integration. - Basic principles and concepts (according to the International Software Testing Qualifications Board - ISTQB) - Test-driven development (TDD) - Functional and non-functional tests - Static and dynamic test methods - Cost-efficient definition of test strategies - Metrics for test management and software quality - Test tools and their introduction in a software development department Optional: - Evaluation of test strategies with CMMI (Capability Maturity Model Integration), SPICE (Software Process Improvement and Capability Determination) and TPI (Test Process Improvement)			
3	Teaching and learning methods: 2 SWS Seminar 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Programming skills in C, basic knowledge of software engineering			

6	Form of the examination: Term paper or written exam
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

AI5117 Master's Thesis in Global Software Development				
Module code FB:		English module name: Master's Thesis in Global Software Development		
Workload: 750 h	ECTS points: 24+ 1 (homework + colloquium)	Semester of study: GSD 2020: 3rd/4th semester	Frequency of the course: Winter and summer semesters	Duration: 1 semester
Art: GSD: compulsory module		Level: Master	Applicability of the module:	
1	Qualification goals: The Master's thesis is a written examination paper. The thesis should show that students are able to independently work on a problem from the field of Global Software Development using scientific methods and present it logically within a specified period of time.			
2	Contents of the module Depending on the respective subject area			
3	Teaching and learning methods: A professor at Fulda University of Applied Sciences will supervise the Bachelor's thesis in terms of subject and methodology.			
4	Language: German or in another language after consultation with the supervising professor			
5	Prerequisites for participation in the module: necessary: 55 ECTS of the modules to be completed in the first and second semesters according to the curriculum; and Master internship, if this module was selected as part of the four-semester course variant recommended: Successful completion of all modules of the 1st and 2nd semester			
6	Form of the examination: Term paper (final thesis in the form of the Master's thesis) and colloquium			
7	Valuation methods: Master's thesis: graded Colloquium: ungraded			
8	Requirements for the awarding of ECTS credits: Passed module examinations (colloquium and term paper)			
9	Remarks: The colloquium is after submission of the Master's thesis. In cases where the would fall within the administrative period of the following semester, it can also be completed during the Master's thesis processing period with the consent of the supervising professor.			

SK5900 Intercultural Communication and Scientific Work				
Module code FB:		English module name: Intercultural Communication and Scientific Work		
Workload: 150 h, thereof 72 h attendance time 78 h self-study		ECTS points: 5 ECTS	Semester of study: DS 2024: 3rd semester GSD 2020: 3rd/4th semester	Frequency of the course: Winter or summer semester
Art: DS, GSD: compulsory module		Level: Master	Applicability of the module:	
1	Qualification goals: Students are of the relevance of cultural differences in the conception, organization and processes of communication. They are familiar with current theories of intercultural communication and are able to reflect on their own cultural framework as well as the cognitive and emotional backgrounds when dealing with other people. They are able to deal with cultural differences in a productive and appreciative way. They can analyze the meaning of proper scientific work and transfer the methods to their own work.			
2	Contents of the module Relevance of intercultural communication in modern theories of intercultural communication • Training concepts for the development of intercultural competence • Cultural differences in work-related forms of communication (negotiation, presentation, moderation, customer-expert relationship) • In-depth scientific work, discussion and presentation • Dealing with scientific literature • Citation technique and significance of plagiarism			
3	Teaching and learning methods: 4 SWS exercise (possibly with online parts)			
4	Language: English			
5	Prerequisites for participation in the module: necessary: none recommended: none			
6	Form of the examination: Portfolio or presentation			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

Compulsory elective modules:

SK5901 Intercultural Project Management				
Module code FB:		English module name: Intercultural Project Management		
Workload: 150 h, thereof 72 h attendance time 78 h self-study		ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: Summer semester
Duration: 1 semester				
Art: GSD: Compulsory elective module		Level: Master	Applicability of the module:	
1	Qualification goals: Students are aware of the particular socio-emotional challenges associated intercultural project management (IPM). They are familiar with the relevant theoretical concepts of IPM. They are able to utilize the advantages of IPM and deal with the associated problems.			
2	Contents of the module Cultural effects of • Team development • Team dynamics • Guidance • Knowledge management • Team performance • virtual teams • Roles in teams and the specific application of IPM in software development			
3	Teaching and learning methods: 2 SWS Seminar 2 SWS Exercise			
4	Language: English			
5	Prerequisites for participation in the module: necessary: none recommended: Knowledge of project management			
6	Form of the examination: Project work or presentation			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5094 Software Development Project				
Module code FB:		English module name: Software Development Project		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: Summer semester	Duration: 1 semester
Art: GSD: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: Students are able to solve a complex software development task in a team using modern project management methods. They show that they can apply their existing knowledge and acquire more in-depth knowledge independently. In , are able to reflect on the methodological approach and the process they have chosen. By working in teams, they improve their communication skills and strategic decision-making and learn to take responsibility.			
2	Contents of the module The task of the project is to take into account the content of the other modules (of the Master's program) so that the students deepen their understanding and see the individual contents in a larger context. The students organize themselves into project teams of 7-10 people. The teams decide on their internal organization and processes. They use the Capability Maturity Model (CMM) to monitor and evaluate their processes.			
3	Teaching and learning methods: 4 SWS Internship Accompanied practical work			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Project Management and Software Engineering			
6	Form of the examination: Unit			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5095 Current Topic of Global Software Development				
Module code FB:		English module name: Current Topic of Global Software Development		
Workload: 150 h, thereof 72 h attendance time 78 h self-study		ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: Summer semester
Art: GSD: Compulsory elective module		Level: Master	Applicability of the module:	
1	Qualification goals: Students expand their knowledge in a current subject area of their field of study and can apply their knowledge in practice.			
2	Contents of the module Depending on the chosen topic, will be announced before the start of the semester			
3	Teaching and learning methods: 2 SWS Seminar 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Depending on the chosen topic, will be announced before the start of the semester			
6	Form of the examination: Project work or written exam or presentation			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5016 Research Project				
Module code FB:		English module name: Research Project		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: AIM 2017 & 2023, DS 2024: 3rd semester GSD 2020: 1st semester	Frequency of the course: AIM, DS: Winter and summer semester	Duration: 1 semester
Art: AIM: Compulsory module DS: Compulsory elective module of the four semesters Study program variant GSD: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: The students • apply strategies of scientific research and assessment of the quality of scientific results in the context of a specific research objective. • derive scientifically comprehensible research questions or hypotheses for a given problem. • define their research project independently. • reflect on and write about their own work process and interim results in their own words and draw corrective conclusions with regard to the achievement of the research objective. • select and apply those methods from the established methods of a research area that are suitable for achieving the specific research (sub)objectives. • can objectively describe, interpret and critically scrutinize the results of their own and external research work.			
2	Contents of the module Students work on a project whose content is chosen in such a way that research questions are further deepened and placed in a larger context. The supervising teacher provides the respective students with guidance on the project content and supports them in acquiring additional, in-depth scientific knowledge and skills required to carry out the project. They also supervise the implementation of the project. Organization and collaboration in project groups is possible, provided that the research topic can be subdivided accordingly and subtasks can be assigned to individual students.			
3	Teaching and learning methods: 4 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: scientific work			
6	Form of the examination: Project work or elaboration			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			

9	Remarks: none
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translation; legally non-binding

AI5122 Advanced Big Data				
Module code FB:		English module name: Advanced Big Data		
Workload: 150 hours, thereof 55 h attendance time 95 h self-study		ECTS points: 5	Semester of study: AIM 2024: 1st/3rd semester AIM 2017: 3rd semester GSD 2020: 1st semester DS 2024: 3rd semester	Frequency of the course: Winter or summer semester
Duration: 1 semester				
Art: GSD: Compulsory elective module AIM 2024: Compulsory elective module (Business Informatics) AIM 2017: Compulsory elective module (data science, business informatics)		Level: Master	Applicability of the module:	
1	Qualification goals: Substantial advances in computing capacities have once again made data and data processing a core topic in computer science. The size and complexity of the data generated in business and industry today require a further developed or modified approach: Big Data and the associated technologies and strategies. After successfully completing this course, students will be able to, - to illustrate the inherent restrictions of established relational database systems. - to work through further applications and use cases in more detail using individual examples and to derive architectural approaches from the big data use cases. - distinguish and apply the basic components of the extended Hadoop ecosystem hands-on. - Assess and evaluate big data / noSQL use cases in the extended Hadoop ecosystem. - the challenges of administering big data solutions in the extended ecosystem. - in particular to evaluate and classify newer and further approaches in the area of big data.			
2	Contents of the module: - Architectural description of big data infrastructures and important application scenarios - Hadoop File System and its components. Applications and exercises - MapReduce and YARN as the basis / essential component of almost all big data infrastructures - Hive, Impala and HCatalog: Data storage and modeling, as well as corresponding applications - Flume as a broker for e.g. large log infrastructures and various implementations - Spark Basics: Systems for main memory-oriented processing - Spark with Scala: Functional programming with Spark (resilient distributed datasets, aggregation, APIs, algorithms) Optional: - recent developments in the area of BigData and special technologies (InfluxDB, Hortworks Sandbox, Cloudera Enterprise Manager, etc.)			
3	Teaching and learning methods:			

	2 SWS Seminar lessons 2 SWS Seminar
4	Language: German or English
5	Prerequisites for participation in the module: necessary: none Recommended: Big Data lecture, Java, Scala (but not mandatory)
6	Form of the examination: Portfolio
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

AI5046 Cryptography - Theory and Applications				
Module code FB:		English module name: Cryptography - Theory and Applications		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: AIM 2017: 2nd/3rd semester GSD /2020: 1st semester	Frequency of the course: Winter or summer semester	Duration: 1 semester
Art: AIM, GSD: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: Students have an overview of cryptography applications in IT systems. In particular, they know the mathematical principles, theories and functions of cryptographic procedures, can evaluate protocols for encryption, authentication and key exchange and implement them in practice for corresponding problems. They are familiar with steganographic methods of covert communication.			
2	Contents of the module • Cryptography, cryptanalysis and steganography • Cryptosystems and classification, Shanonn's theory, factorization problem, discrete logarithm problem • Theory of cryptographic methods (AES, RSA, ElGamal, Digital Signature Algorithm, Elliptic Curve Cryptosystems) • Key management (Session Key Distribution Scheme and Key Agreement Scheme) • Authentication and cryptographic protocols, including on-time password, challenge-response and zero-knowledge protocols • Cryptographically strong hash functions, Merkle-Damgard construction principle, message authentication codes • Theory of steganographic methods • Introduction to steganographic embedding algorithms for various carrier media (images, audio and video) • Current IT security problems			
3	Teaching and learning methods: 4 SWS Seminar lessons			
4	Language: English			
5	Prerequisites for participation in the module: necessary: none recommended: Algebraic foundations of computer science, IT security			
6	Form of the examination: Written exam or technical discussion			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination			
9	Remarks: none			

AI5083 User-Centered Development				
Module code FB:		English module name: User-Centered Development		
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: GSD 2020: 1st semester	Frequency of the course: Summer semester	Duration: 1 semester
Art: GSD 2020: Compulsory elective module	Level: Master	Applicability of the module:		
1	Qualification goals: The students - integrate principles of human-centered design and evaluation into the development of interactive systems. - illustrate the effects of user requirements on design in general and on examples. - explain contextual task analysis and how it can contribute to development. - formulate measurable qualitative and quantitative usability goals and associated criteria. - design a user interface iteratively according to human-centered design methods and evaluate it. - develop prototypes taking into account different interaction modalities and styles - plan, organize and carry out evaluations and analyze and document the results (expert inspections and user studies). Students strengthen their group work skills by collaborating with other team members.			
2	Contents of the module Basics - Human-computer interaction - Usability engineering, human-centered design process especially iterative, participative UI design - Usability principles and guidelines Process - Planning the human-centered process - Understanding and indicating the context of use - Definition of user and organizational requirements - Production of design solutions (prototypes) - Evaluation of the design based on requirements - Embedding usability activities in the software engineering lifecycle Optional: - Tool insert			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Internship			
4	Language: German or English			
5	Prerequisites for participation in the module: necessary: none recommended: Software Engineering			
6	Form of the examination: Portfolio or written exam			

7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination and active participation
9	Remarks: none

AI5174 Stream Processing and Analytics				
	English module name: Stream Processing and Analytics			
Workload: 150 h, thereof 72 h attendance time 78 h self-study	ECTS points: 5 ECTS	Semester of study: AI 2024: 1st/3rd semester DS 2024: 3rd semester GSD 2020: 1st semester	Frequency of the course: Winter or summer semester	Duration: 1 semester
Art: AI, GSD: Compulsory elective module DS: Compulsory elective module of the four-semester study program vari- ante	Level: Master	Applicability of the module:		
1	Qualification goals: The students - name the essential characteristics of a data stream as well as different processing paradigms such as batch and stream processing. - describe the differences between limited and unlimited data streams. - break down unlimited data streams into limited data streams using windowing. - process data streams using common frameworks. - distinguish between processing time and event time and process out-of-order and late-arriving data robustly. - process streaming data according to defined requirements using the various types of streaming joins. - describe the challenges and opportunities of data stream analytics and online machine learning. - the essential characteristics of data sketches. - explain selected data sketches and - them for practical problems. - explain selected online machine learning methods and - them for practical problems. - evaluate the current state of research in the field of stream processing and analytics and - are actively in applied research in this area.			
2	Contents of the module			
	- Basics: data streams, windowing, batch processing, stream processing - Robust stream processing: out-of-order data, triggers, watermarks - Streaming Joins - Basics: stream analytics, online machine learning, data sketches - Selected data sketches - Selected methods of online machine learning			
3	Teaching and learning methods: 2 SWS Seminar lessons 2 SWS Practical course			
4	Language: German or English			

5	Prerequisites for participation in the module: necessary: none recommended: Machine Learning, Big Data Technologies
6	Form of the examination: Project work or portfolio
7	Valuation methods: graded
8	Requirements for the awarding of ECTS credits: Passed module examination
9	Remarks: none

Modules for the four-semester variant:

AI4001 Additional Courses				
Module code FB:		English module name: Additional Courses		
Workload: 900 h,		ECTS points: 30 ECTS	Semester of study: GSD 2020: 3rd/4th semester	Frequency: Winter and summer semesters
Duration: 1 semester				
Art: GSD: Compulsory elective module of the four-semester course variant		Level: Bachelor's degree	Applicability of the module:	
1	Qualification goals: Students expand and deepen their study-related knowledge by taking modules from the degree programs of the Department of Applied Computer Science with regard to the desired degree, see § 5 para. 3. The respective qualification objectives result the module descriptions of the modules taken.			
2	Contents of the module See module descriptions of the selected modules			
3	Teaching and learning methods: See module descriptions of the selected modules			
4	Language: Depending on the module			
5	Prerequisites for participation in the module: necessary: four-semester course variant, DSH-2 recommended: none			
6	Form of the examination: See module descriptions of the selected modules.			
7	Valuation methods: See module descriptions of the selected modules.			
8	Requirements for the awarding of ECTS credits: See module descriptions of the selected modules.			
9	Remarks: The modules are determined in consultation with the program director			

AI5119 Master Internship				
Module code FB:		English module name: Master Internship		
Workload: 800 h internship 100 h self-study		ECTS points: 30 ECTS	Semester of study: GSD 2020: 3rd semester	Frequency: Winter and summer semesters
Art: GSD: Compulsory elective module of the four-semester course variant		Level: Master	Applicability of the module:	
1	Qualification goals: Students are to apply the specialist and methodological knowledge they have learned in a corporate environment. In addition, they should demonstrate their social and intercultural competence in specific projects and learn to adapt to operational conditions.			
2	Contents of the module Depending on the area of work in the company. The area of activity should be selected and specified in the internship contract in such a way that the Master's thesis can also be written in this area.			
3	Teaching and learning methods: Supervised practical phase in the company. The practical phase lasts 20 weeks full-time			
4	Language: German or English			
5	Prerequisites for participation in the module: required: four-semester degree program variant AND proof of at least 40 ECTS of the modules required for the degree recommended: none			
6	Form of the examination: Report			
7	Valuation methods: graded			
8	Requirements for the awarding of ECTS credits: Passed module examination, proof of completed practical phase			
9	Remarks: Further regulations on the Master's internship can be found in the practical regulations for the Master's degree course in Global Software Development.			

AI5118 German Language				
Module code FB:		English module name: German Language		
Workload: 900 h	ECTS points: 30 ECTS	Semester of study: GSD 2020: 3rd/4th se- mester	Frequency of the course: Winter and summer semester	Duration: 1 semester
Art: GSD: Compulsory elective module of the four-semester course variant	Level:	Applicability of the module:		
1	Qualification goals: Students have acquired language skills in German after completing the relevant modules, see §5 Para. 5			
2	Contents of the module By offer			
3	Teaching and learning methods: By offer			
4	Language: German			
5	Prerequisites for participation in the module: Necessary: four-semester course variant recommended: none			
6	Form of the examination: By offer			
7	Valuation methods: By offer			
8	Requirements for the awarding of ECTS credits: By offer			
9	Remarks: German courses are determined and selected in consultation with the head of degree program.			

AI5092 Intercultural Study of Computer Science				
Module code FB:		English module name: Intercultural Study of Computer Science		
Workload: 900 h		ECTS points: 30 ECTS	Semester of study: GSD 2020: 3rd/4th semester	Frequency: Winter and summer semesters
Art: GSD: Compulsory elective module of the four-semester course variant		Level:	Applicability of the module:	
1	Qualification goals: Students deepen their knowledge of computer science and learn how this discipline is taught and understood in other cultures. They experience different teaching and examination styles. They have a wealth of experience in dealing with everyday situations and communication attitudes in an intercultural environment. They can reflect on their experiences during their stay abroad.			
2	Contents of the module Students spend a semester at a university in a non-German-speaking country. They attend the courses of a Bachelor's program (in the second half of their studies) and take the corresponding examinations.			
3	Teaching and learning methods: The teaching and learning methods depend on the modules of the foreign university			
4	Language: depending on the language of instruction at the foreign university			
5	Prerequisites for participation in the module: necessary: Students on the four-semester degree program variant are required to study abroad accordance with § 6 Para. 1 recommended: none			
6	Form of the examination: The forms of examination depend on the modules of the foreign university			
7	Valuation methods: ungraded			
8	Requirements for the awarding of ECTS credits: Passed module examinations; written report reflecting on the experience			
9	Remarks: The foreign university and the modules to be taken are selected in consultation with the head of the degree program.			

Appendix 3: Professional practice regulations

Practical work regulations for the Master's degree program in Global Software Development

§ 1 General

- (1) In accordance with § 5 Para. 2 (b) of the examination regulations for the Master's degree course in Global Software Development, students can complete a Master's internship in the four-semester course variant. It is prepared and supervised by the university.
- (2) If students have a degree from a German university as a prerequisite for admission to the course, the Master's internship is completed as an internship at a company, organization or research institution abroad.
- (3) If students have a degree from a foreign university as a prerequisite for admission to the course, the Master's internship is completed as an internship at a company, organization or research institution in Germany.
- (4) The Master's internship is regulated on the basis of a contract between the student and the practical placement.

§ 2 Objectives and tasks

The aims of the Master Internship are to familiarize students with German professional practice and to acquire practical skills by working on tasks in the field of software development.

§ 3 Status of students

- (1) During the Master's internship, students remain enrolled at HS Fulda with all rights and obligations.
- (2) The students are not interns within the meaning of the Vocational Training Act and are not subject to the Works Constitution Act or the Employee Representation Act during the Master Internship.
- (3) They are obliged to comply with the instructions of the practice location and the person appointed by it that are necessary to achieve the objectives and to observe the regulations applicable to the practice location, in particular work regulations and accident prevention regulations as well as regulations on confidentiality.

§ 4 Duration and timing

- (1) The Master Internship comprises a continuous period of twenty weeks at a practice location. Interruptions must always be made up for.
- (2) The Master's internship should take place in the third semester.
- (3) The daily working hours correspond to the usual working hours of a full-time position within the practice, but not less than 35 hours per week.

§ 5 Admission

Students who have demonstrably completed modules amounting to 40 ECTS of the modules required for the degree are admitted to the Master's internship.

§ 6 Practice location

- (1) The practical placements are usually proposed by the students. If the proposal cannot be approved, the department should arrange an internship.
- (2) Students should be supervised at the work placement by a person appointed by the work placement location who has appropriate training in a relevant subject area and full-time at the work placement location. The supervising person has the task of regulating and monitoring the instruction of the students in the areas of work and tasks.

§ 7 Supervision by the university

- (1) The Practice Department of the Faculty of Applied Computer Science advises students primarily on formal issues. This includes in particular
 - (a) the selection and recognition of internships,
 - (b) the review and confirmation of contracts,
 - (c) the evaluation and verification of the proper completion of the Master-In-ternship,
 - (d) Advice in the event of conflicts between students and supervisors in the partner companies.
- (2) A member of the Department of Applied Computer Science supervises and advises students in all technical matters relating to the Master's internship.
- (3) Students are obliged to inform the university supervisor in detail about the progress of their work at the end of each month of the Master's internship.

§ 8 Contract

- (1) Before the start of the Master's internship, students a contract with company offering the Master's internship. The contract must be submitted to the internship department for approval. The Master internship must be registered before the start of the internship.
- (2) The contract regulates in particular
 - (a) the commitment of the students,
 - comply with the instructions of the practice center and the persons commissioned by it,
 - carry out the assigned tasks carefully,
 - the regulations applicable to the practice location, in particular
 - work regulations and accident prevention regulations as well as regulations on the duty of confidentiality,
 - prepare a chronologically structured report in accordance with the requirements of the faculty, showing the progress of the practical training (report on the Master's internship).

(b) the obligation of the practice location,

- carefully train the students in accordance with the professional practice regulations,
- to draw up a work plan in coordination with the departmental supervisor,
- issue the student with a qualified report on the progress and content of the Master's internship and check and sign the report to be drawn up,
- to enable students to take examinations in the Department of Applied Informatics,
- to nominate a person as a practical supervisor and as a contact person at Fulda University of Applied Sciences.

§ 9 Recognition

- (1) Students apply for recognition of the Master's internship by submitting the report approved by the departmental supervisor and proof of activity.
- (2) If the Master Internship is recognized, the Master Internship module receives the assessment "participated with success".